

Hypertension: Definition and Classification

Hypertension is persistently elevated arterial blood pressure. Under the 2017 ACC/AHA guideline, normal blood pressure is below 120/80 mmHg. Elevated blood pressure is a systolic pressure of 120-129 mmHg with a diastolic pressure below 80 mmHg. Stage 1 hypertension is 130-139 mmHg systolic or 80-89 mmHg diastolic. Stage 2 hypertension is 140/90 mmHg or higher. Diagnosis should be based on an average of two or more readings obtained on two or more occasions. Home or ambulatory monitoring helps exclude white coat hypertension.

Lifestyle Management

Lifestyle changes are recommended for every patient with elevated blood pressure or hypertension. The DASH diet, rich in fruits, vegetables and low-fat dairy, lowers systolic pressure by about 11 mmHg in hypertensive adults. Sodium intake should ideally be below 1500 mg per day. Other measures include weight loss, 90 to 150 minutes of aerobic exercise per week, limiting alcohol to no more than two drinks per day for men and one for women, and increasing dietary potassium.

Drug Treatment and Hypertensive Crisis

First-line antihypertensive drug classes are thiazide diuretics, ACE inhibitors, angiotensin receptor blockers (ARBs) and calcium channel blockers. ACE inhibitors such as lisinopril can cause a persistent dry cough in up to 10 percent of patients; switching to an ARB such as losartan usually resolves it. ACE inhibitors and ARBs are contraindicated in pregnancy. A hypertensive crisis is a blood pressure above 180/120 mmHg. When accompanied by acute target organ damage it is a hypertensive emergency requiring admission and intravenous therapy.